Unit 2 practice solutions

Worked answers to all 13 questions in the supplied practice-session photo.

The supplied data sheet mentioned on the sheet was not attached. The numerical bond energies below come from Question 6; the other solutions use standard bonding and acid-base models. Dots in Lewis diagrams represent lone pairs.

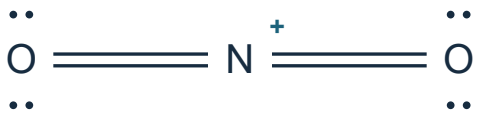
1. Lewis structures and formal charges

Formal charge (FC) = valence electrons - nonbonding electrons - ½(bonding electrons)

(a) NO_2^+: nitronium ion

1. Count valence electrons: 5 + 2(6) - 1 = **16 electrons**.

2. Connect O-N-O and make two N=O double bonds. Give each O two lone pairs. All three atoms now have octets; N has no lone pairs.



Each O: 2 lone pairs; N: 0 lone pairs

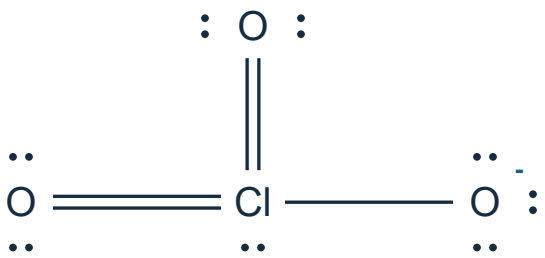
3. FC(N) = 5 - 0 - 4 = **+1**. Each O: 6 - 4 - 2 = **0**. The formal charges add to +1.

Expanded central-atom octet? No. Nitrogen has exactly 8 electrons around it.

(b) ClO_3^-: chlorate ion

1. Count valence electrons: 7 + 3(6) + 1 = **26 electrons**. Three Cl-O single bonds and complete O octets leave one lone pair on Cl.

2. That octet-only drawing has Cl = +2 and three O = -1. In the usual expanded-octet Lewis treatment, convert two O lone pairs into Cl=O bonds to reduce charge separation:



One of three equivalent major contributors

3. Cl: 7 - 2 - 5 = **0**; each double-bonded O: 6 - 4 - 2 = **0**; single-bonded O: 6 - 6 - 1 = **-1**. Cl has one lone pair; the double-bonded O atoms have two each; the single-bonded O has three.

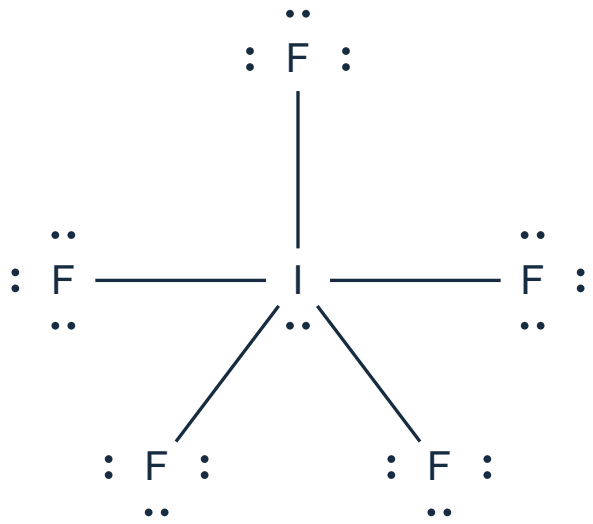
Expanded central-atom octet? Yes in this major-contributor drawing: Cl has 12 electrons around it. There are three equivalent placements of the single bond.

1. Lewis structures, continued

(c) IF_5: iodine pentafluoride

1. Count valence electrons: 7 + 5(7) = **42 electrons**.

2. Five I-F single bonds use 10 electrons. The three lone pairs on each of the five F atoms use another 30 electrons. Put the remaining 2 electrons on I as one lone pair.



Connectivity sketch: 5 I-F bonds and 1 lone pair on I.

3. FC(I) = 7 - 2 - 5 = **0**. Each F: 7 - 6 - 1 = **0**. The total is zero. The sketch shows connectivity; VSEPR gives a square-pyramidal molecular shape.

Expanded central-atom octet? Yes. I has 5 bonding pairs + 1 lone pair = 12 electrons.

(d) CO: carbon monoxide

1. Count valence electrons: 4 + 6 = **10 electrons**.

2. A C≡O triple bond uses 6 electrons; place the remaining 4 as one lone pair on each atom. Both atoms have octets.



One lone pair on each atom; overall charge = 0

3. FC(C) = 4 - 2 - 3 = **-1**. FC(O) = 6 - 2 - 3 = **+1**. The charges sum to zero. These are formal charges, not measured partial charges.

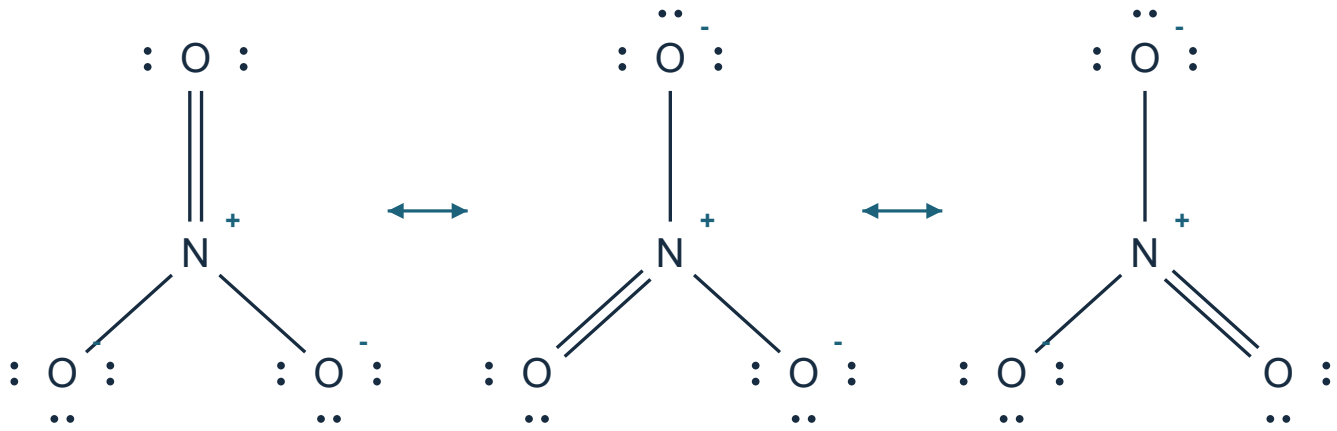
Expanded octet? No. CO is diatomic and each atom has exactly 8 electrons around it.

Expanded octets here describe the conventional Lewis bookkeeping model. They do not establish that central-atom d orbitals form the bonds.

2. Resonance and measured bond lengths

(a) NO_3^-: three equivalent major contributors

1. Count 5 + 3(6) + 1 = **24 valence electrons**. Keep the N and O nuclei fixed. Put one N=O double bond in each possible position.



2. In each contributor: N has **FC = +1**; the two single-bonded O atoms each have **FC = -1**; the double-bonded O has FC = 0. N has no lone pair, each single-bonded O has three, and the double-bonded O has two.

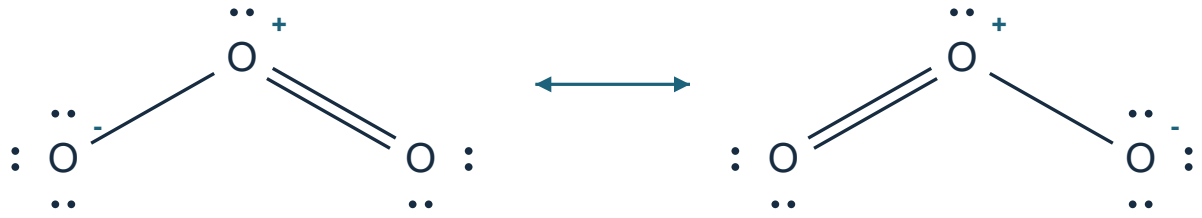
3. Every contributor has an octet on N. A drawing with two N=O double bonds would exceed nitrogen's octet and is not an allowed Lewis contributor.

Average N-O bond order = (2 + 1 + 1) / 3 = **4/3**

All three N-O bond lengths are equal, intermediate between typical N-O single and double bonds.

(b) O_3: two equivalent major contributors

1. Count 3(6) = **18 valence electrons**. Use one O-O single bond and one O=O double bond; move the double bond to obtain the second contributor.



Central O: 1 lone pair in each contributor

2. Central O: 6 - 2 - 3 = **+1**. Single-bonded terminal O: 6 - 6 - 1 = **-1**. Double-bonded terminal O: 6 - 4 - 2 = **0**.

Average O-O bond order = (1 + 2) / 2 = **1.5**

The two O-O bonds have equal lengths, intermediate between typical O-O single and double bonds.

The measured species is a resonance hybrid with delocalized electrons. It does not alternate between separate molecules with different bond lengths.

3. VSEPR geometry and polarity

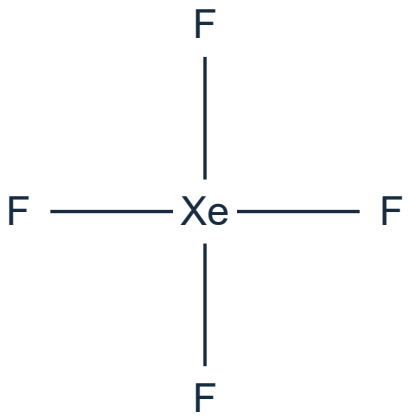
1. Count valence electrons, complete the F octets, and put any remaining electrons on the central atom. Each bond counts as one electron domain.

XeF_4: 8 + 4(7) = 36 e^-; 36 - 8 (bonds) - 24 (F lone pairs) = 4 e^- on Xe

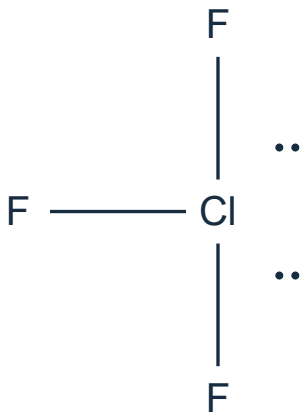
ClF_3: 7 + 3(7) = 28 e^-; 28 - 6 (bonds) - 18 (F lone pairs) = 4 e^- on Cl

2. Four remaining electrons give **two central-atom lone pairs** in each molecule. Count bonding domains plus lone pairs to assign the electron-domain geometry.

Property	XeF_4	ClF_3
VSEPR type	AX_4E_2	AX_3E_2
Electron domains	6 = 4 bonds + 2 lone pairs	5 = 3 bonds + 2 lone pairs
Electron-domain geometry	Octahedral	Trigonal bipyramidal
Lone-pair arrangement	Opposite (trans) positions	Two equatorial positions
Molecular geometry	**Square planar**	**T-shaped**
Molecular polarity	**Nonpolar**	**Polar**



XeF_4: square planar



ClF_3: T-shaped

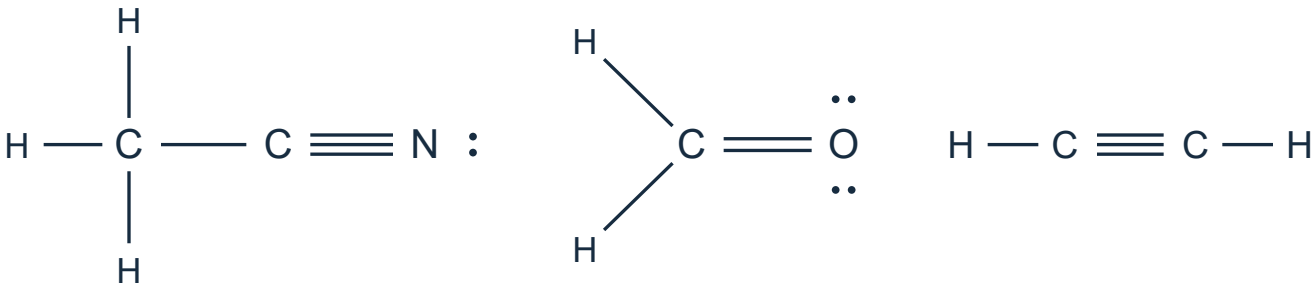
In the XeF_4 sketch, its two lone pairs lie on opposite sides of the page, perpendicular to the square of F atoms. In the ClF_3 sketch, the two dot pairs indicate its equatorial lone pairs.

3. Justify the polarity. In square-planar XeF_4, equal Xe-F bond dipoles cancel in opposite pairs. In ClF_3, the two approximately opposed axial bond dipoles largely cancel, but the equatorial Cl-F dipole has no opposing partner. The molecule therefore has a net dipole.

XeF_4: nonpolar square planar. ClF_3: polar T-shaped.

4. Carbon hybridization and sigma / pi bonds

1. Draw the connectivity. A single bond contains **1 sigma (σ)**; a double bond contains **1 σ + 1 pi (π)**; a triple bond contains **1 σ + 2 π**.



(a) CH_3CN (b) CH_2O (c) C_2H_2

2. At each carbon, count electron domains. A multiple bond is **one domain**: 4 domains imply sp^3, 3 imply sp^2, and 2 imply sp.

Molecule	Carbon hybridization and local shape	Total σ	Total π
(a) CH_3CN	CH_3 carbon: **sp^3**, tetrahedral. C≡N carbon: **sp**, linear.	5	2
(b) CH_2O	Carbonyl carbon: **sp^2**, trigonal planar.	3	1
(c) C_2H_2	Both carbons: **sp**, linear.	3	2

(a) CH_3CN: H_3C-C≡N

3. The methyl carbon has three C-H bonds and one C-C bond: four domains. The nitrile carbon has one C-C bond and one C≡N bond: two domains.

σ = 3(C-H) + 1(C-C) + 1(C≡N) = **5**; π = 2(C≡N) = **2**

(b) CH_2O: formaldehyde

4. The carbon has two C-H bonds and one C=O bond: three domains, so sp^2 with angles near 120°. The O has two lone pairs.

σ = 2(C-H) + 1(C=O) = **3**; π = 1(C=O) = **1**

(c) C_2H_2: acetylene

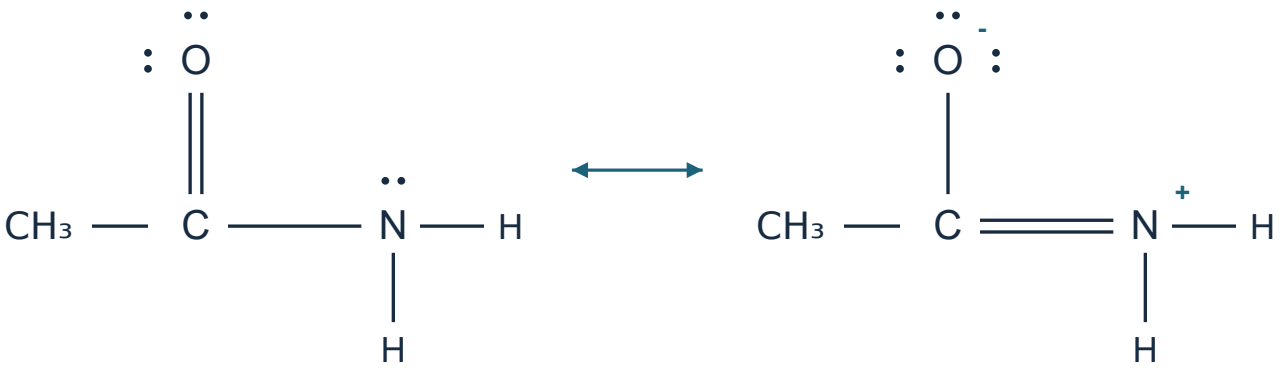
5. Each carbon has one C-H bond and one C≡C bond: two domains. Both carbons are sp and the H-C-C-H skeleton is linear (180°).

σ = 2(C-H) + 1(C≡C) = **3**; π = 2(C≡C) = **2**

Count one σ for every connected atom pair, then add one π for each double bond and two π for each triple bond.

5. Acetamide: hybridization and planarity

Structure: $CH_3C(=O)NH_2$. The two drawings below keep the same atoms connected and move only electrons.



Atom	Hybridization	Local shape / explanation
Methyl carbon, CH_3	sp^3	Tetrahedral; four σ bonds.
Carbonyl carbon, C=O	sp^2	Trigonal planar; three σ directions and a p orbital for π bonding.
Amide nitrogen	Approximately sp^2	Nearly trigonal planar; three σ bonds, with its lone pair delocalized in a p orbital.

1. The N lone pair overlaps with the carbonyl π system. The charge-separated contributor has O^-, a C=N double bond, and N^+.

2. Effective p-orbital overlap favors a planar O-C-N unit. The C-N bond has partial double-bond character, so rotation is restricted. An ordinary amine has a more localized lone pair and is usually pyramidal.

A lone-pair count on just the neutral contributor can suggest sp^3 N. For an amide, the resonance-aware basic bonding model uses approximately sp^2 N.

6. Estimating the C≡C bond energy

The target bond symbol is faint in the photograph. The main answer below interprets it as a C≡C triple bond. If the intended target is the C-C single bond, its energy is already given as 347 kJ mol^{-1}.

1. Use the given energies: C-C = 347 and C=C = 614 kJ mol^{-1}. The first additional π interaction raises the energy by 614 - 347 = **267 kJ mol^{-1}**.

2. A triple bond should be stronger than a double bond, but it is not three identical single bonds. Of the choices, **839** is the chemically reasonable triple-bond estimate. It corresponds to another increment of 839 - 614 = **225 kJ mol^{-1}**.

Best listed C≡C estimate: 839 kJ mol^{-1}. The two given values alone do not determine a unique numerical triple-bond energy.

3. A triple bond is one σ plus two π interactions; π overlap differs from σ overlap, and bonding changes with hybridization and molecular environment. Thus 3 × 347 = 1041 treats unlike interactions as equal and overestimates the listed value.

Average energies may still be summed over *different bonds* when estimating a reaction enthalpy. The mistake here is assigning identical, independent energy increments to the components of a multiple bond.

7. Molecular orbitals, bond order, and magnetism

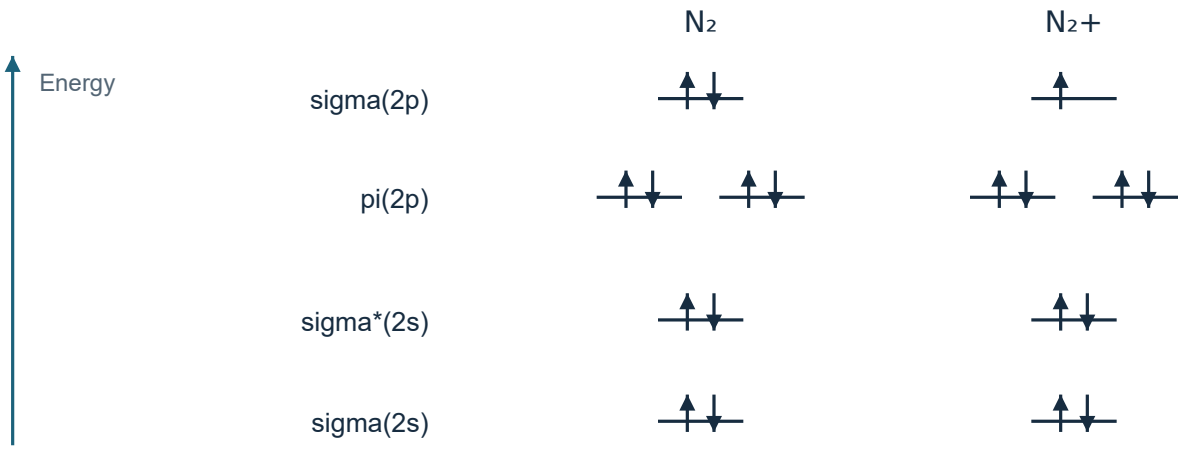
Bond order = ½(number of bonding electrons - number of antibonding electrons)

(A) N_2 and N_2^+

1. Each N contributes 5 valence electrons: N_2 has **10**, and N_2^+ has **9**. For N_2, the bonding π(2p) orbitals lie below the bonding σ(2p) orbital.

N_2: $[\sigma(2s)]^2[\sigma^*(2s)]^2[\pi(2p)]^4[\sigma(2p)]^2$

N_2^+: $[\sigma(2s)]^2[\sigma^*(2s)]^2[\pi(2p)]^4[\sigma(2p)]^1$



Only the occupied valence MO levels are drawn. The two π orbitals are degenerate; each holds two electrons. Core orbitals are omitted because their bonding and antibonding contributions cancel.

2. N_2: bonding electrons = 2 + 4 + 2 = **8**; antibonding electrons = **2**. Bond order = (8 - 2)/2 = **3**.

3. Ionization removes one electron from the highest occupied σ(2p) **bonding** orbital. N_2^+: bond order = (7 - 2)/2 = **2.5**.

Species	Bond order	Unpaired electrons	Magnetism
N_2	3	0	Diamagnetic
N_2^+	2.5	1	Paramagnetic

Bond strength: $N_2 > N_2^+$. Bond length: $N_2 < N_2^+$. Only N_2^+ is paramagnetic.

(B) Be_2 in the simple valence MO model

1. Each Be is $2s^2$, so the pair supplies 4 valence electrons. Fill $[\sigma(2s)]^2[\sigma^*(2s)]^2$.

2. Bond order = (2 - 2)/2 = **0**. Equal bonding and antibonding occupancies cancel, so this simple model predicts no stable net bond.

8. Brønsted acids, bases, and amphiprotic species

1. A Brønsted acid donates H^+; a Brønsted base accepts H^+. An amphiprotic species can do both in suitable reactions.

Species	Classification	Reason
HSO_4^-	**Amphiprotic**	Can lose H^+ to form SO_4^{2-} or gain H^+ to form H_2SO_4.
NH_4^+	**Acid**	Donates H^+ to form NH_3.
CO_3^{2-}	**Base**	Accepts H^+ to form HCO_3^-; has no hydrogen to donate.
H_2O	**Amphiprotic**	Can donate H^+ to form OH^- or accept H^+ to form H_3O^+.

HSO_4^-: show both roles

2. As an acid, it transfers its proton to water:

$$HSO_4^- + H_2O \rightleftharpoons SO_4^{2-} + H_3O^+$$

As a base, it can accept a proton from hydronium:

$$HSO_4^- + H_3O^+ \rightleftharpoons H_2SO_4 + H_2O$$

The second equation illustrates proton acceptance; it is unfavorable in dilute water because H_2SO_4 strongly ionizes. HSO_4^- mainly behaves as an acid in water.

NH_4^+ and CO_3^{2-}

3. Transfer one proton, then verify both atom counts and total charge:

$$NH_4^+ + H_2O \rightleftharpoons NH_3 + H_3O^+$$

$$CO_3^{2-} + H_2O \rightleftharpoons HCO_3^- + OH^-$$

H_2O: show both roles

4. As an acid, water donates a proton to ammonia:

$$H_2O + NH_3 \rightleftharpoons OH^- + NH_4^+$$

As a base, water accepts a proton from HCl:

$$H_2O + HCl \rightarrow H_3O^+ + Cl^-$$

Each conjugate acid-base pair differs by exactly one H^+. Its charge changes by exactly one unit.

9. Ammonia as a Brønsted and Lewis base

(a) Brønsted base in water

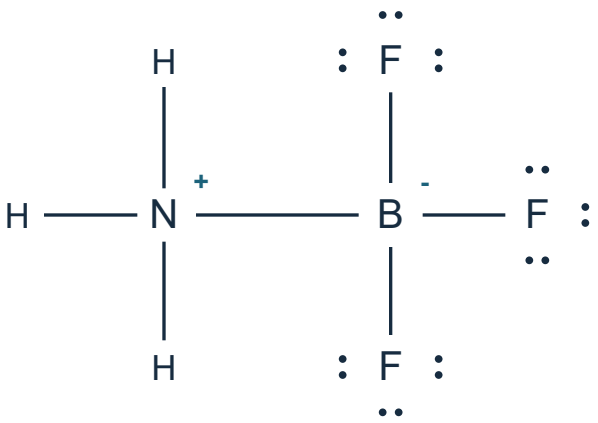
$$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$$

1. NH_3 accepts H^+, so it is the **Brønsted base**; H_2O donates H^+, so it is the acid. NH_4^+ and OH^- are the conjugate acid and base.

(b) Lewis base with BF_3

$$H_3N: + BF_3 \rightarrow H_3N{\rightarrow}BF_3$$

2. The N lone pair supplies both electrons of the new N-B bond. **N in NH_3 is the electron-pair donor (Lewis base)**; **B in BF_3 is the electron-pair acceptor (Lewis acid)**. B begins with only six electrons around it and reaches an octet in the adduct.



Formal charges in the adduct: N = +1; B = -1; net = 0

The coordinate-bond arrow points from N to B. After formation, the N-B bond is an ordinary two-electron covalent bond; no proton transfer is needed.

10. Why phosphorous acid is diprotic

1. Use the stated connectivity, **$H\text{-}P(=O)(OH)_2$**. Two H atoms are bonded to O; the third is bonded directly to P.

2. Remove one O-H proton at a time. Both ionizations retain the P-H bond:

$$HP(=O)(OH)_2 + H_2O \rightleftharpoons HP(=O)(OH)(O^-) + H_3O^+$$

$$HP(=O)(OH)(O^-) + H_2O \rightleftharpoons HP(=O)(O^-)_2 + H_3O^+$$

The same equations written with molecular formulas are:

$$H_3PO_3 + H_2O \rightleftharpoons H_2PO_3^- + H_3O^+$$

$$H_2PO_3^- + H_2O \rightleftharpoons HPO_3^{2-} + H_3O^+$$

H_3PO_3 is diprotic because only its two O-H hydrogens ionize in ordinary aqueous acid-base reactions. Its P-H hydrogen is retained.

11. Hydrolysis, hydration, and acidity

(A) PCl_5 with excess water

1. Write H_3PO_4 and HCl as products. One P fixes one H_3PO_4; five Cl require five HCl; four O require four H_2O.

$$\mathbf{PCl_5 + 4\ H_2O \rightarrow H_3PO_4 + 5\ HCl}$$

Check: P = 1, Cl = 5, O = 4, and H = 8 on each side. Excess water allows complete hydrolysis. The stages can be written as:

$$PCl_5 + H_2O \rightarrow POCl_3 + 2\ HCl$$

$$POCl_3 + 3\ H_2O \rightarrow H_3PO_4 + 3\ HCl$$

(B) SO_3 with water

2. Hydrate this nonmetal oxide to form sulfuric acid:

$$\mathbf{SO_3 + H_2O \rightarrow H_2SO_4}$$

(C) Are the resulting solutions acidic?

3. Yes, both. The covalent nonmetal chloride PCl_5 hydrolyzes to HCl and H_3PO_4. HCl produces H_3O^+ in water; phosphoric acid also donates protons. SO_3 is an **acidic oxide** (the anhydride of H_2SO_4), so hydration yields an acid.

$$HCl + H_2O \rightarrow H_3O^+ + Cl^-$$

$$H_2SO_4 + H_2O \rightarrow H_3O^+ + HSO_4^-$$

12. Predicting the stronger acid from structure

(a) $HClO_2$ versus $HClO_3$

1. The central atom is the same. $HClO_3$ has one more oxygen that is not part of the O-H group. The extra oxygen withdraws electron density and helps stabilize the conjugate base through charge delocalization.

$HClO_3$ is stronger than $HClO_2$. For oxyacids of the same central atom, more terminal O atoms usually increase acidity.

(b) H_2S versus H_2Se

2. Move down Group 16 from S to Se. Se is larger and the H-Se bond is weaker than H-S, making proton loss easier. The larger conjugate-base anion also spreads its charge over a larger volume.

H_2Se is stronger than H_2S. Down this group, weaker H-E bonding outweighs the decrease in electronegativity.

13. Oxidation states and oxide character

(A) Sulfur oxidation states

1. Assign F = -1, ordinary O = -2, and H = +1 when bonded to a nonmetal. The oxidation states in a neutral molecule add to zero. Let x be the sulfur oxidation state.

Compound	Equation	Sulfur oxidation state
SF_4	x + 4(-1) = 0	**+4**
SO_3	x + 3(-2) = 0	**+6**
H_2S	2(+1) + x = 0	**-2**

Oxidation state and formal charge are different bookkeeping systems. Oxidation state assigns bond electrons to the more electronegative atom; formal charge divides bond electrons equally.

(B) MgO and SO_3 with water

2. MgO is a basic oxide. It is a largely ionic metal oxide. Its oxide component accepts protons from water, and hydration gives magnesium hydroxide:

$$\mathbf{MgO(s) + H_2O(l) \rightarrow Mg(OH)_2(s)}$$

$$Mg(OH)_2(s) \rightleftharpoons Mg^{2+}(aq) + 2\ OH^-(aq)$$

Hydration can be slow, and $Mg(OH)_2$ is sparingly soluble. The dissolved portion supplies OH^-, so the aqueous phase is basic.

3. SO_3 is an acidic oxide. It is a covalent nonmetal oxide with sulfur in the +6 oxidation state. Water addition produces sulfuric acid:

$$\mathbf{SO_3 + H_2O \rightarrow H_2SO_4}$$

$$H_2SO_4 + H_2O \rightarrow H_3O^+ + HSO_4^-$$

Oxide	Element character	Behavior with water
MgO	Mg is metallic; oxide is largely ionic.	Forms a hydroxide; aqueous phase is **basic**.
SO_3	S is nonmetallic; oxide is covalent.	Forms an oxyacid; solution is **acidic**.

For these examples: metal oxide → basic hydroxide; nonmetal oxide → oxyacid. The reaction products explain the observed acidity or basicity.